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## Charge Delocalization on *BO* Centers to Improve Conductivity on Single Lithium Ion Conducting Polymer Electrolytes: A Computational/Experimental Approach

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**Charge Delocalization on  $BO_4^-$  Centers to Improve  
Conductivity on Single Lithium Ion Conducting Polymer  
Electrolytes: A Computational/Experimental Approach**

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## ABSTRACT

Single lithium ion conducting polymer electrolytes (SLICPE) could be one of the best options for safer lithium-ion batteries, only if new strategies can be envisaged to improve their ionic conduction properties. In this work, we examine the effect of two strategies to improve the ionic conductivity of SLICPE based on  $sp^3$  boron atoms linkers by PEG chains, namely partial cross-linking and modification of the electron-withdrawing properties of the substituents attached to the  $BO_4^-$  groups. The molecular structure analysis, determined by Density Functional Theory, shows an important variation on the electron density of the  $BO_4^-$  group with presence of electron-withdrawing groups, allowing lower  $Li^+$ -polymer interaction energy, and lower activation energy for  $Li^+$  hopping; these results confirm the changes in the chemical environment of B and Li established by NMR. The increase of the electron-withdrawing capacity of the substituent groups on the  $BO_4^-$  groups decreases their electronic density, increases the length of the  $BO_4^- - Li^+$  bond distance and allows higher mobility of  $Li^+$  in the partially cross-linked SLICPE structure. The ionic conductivity measurements obtained by EIS reveal that the effect of the diminution of electron density in the  $BO_4^-$  groups and its effect on the mobility of  $Li^+$  in the polymer matrix, is mainly observed at temperatures higher than  $T_m$ , where the system has enough energy to break electrostatic short-range interactions within the polymer matrix which limits the mobility of the polymer matrix.

## 1. Introduction

Liquid electrolytes commonly utilized for Lithium-Ion batteries (LIB) are responsible for most of the safety and toxicological issues on the operation of high energy batteries <sup>1</sup>. When LIB are overheated due to overcharging, internal short-circuit, manufacturing defects, physical damage or other failure mechanisms, the organic liquid solutions in LIB generate highly flammable compounds which will directly cause catastrophic thermal accidents <sup>2</sup>. Due to the expected applications of large capacity LIB modules in EVs and smart grids <sup>3</sup>, the quest for safer and more reliable electrolyte systems is therefore urgent. One of the systems with highly desirable properties are Solid Polymer Electrolytes (SPE) <sup>4</sup>.

Conventional SPEs for LIBs are composed of lithium salts dissolved in a polymer matrix in which both, anions and cations are mobile; the presence of both ions involves to the formation of salt concentration gradients and cell polarization during battery operation <sup>5,6</sup>. One of the best solutions to these problems is the immobilization of anions within the polymeric matrices either by the formation of simple electrostatic interactions or covalent bonds; the immobilization of the anion results in the formation of the so-called single lithium ion conducting polymer electrolytes (SLICPE) <sup>6-8</sup>. SLICPE generally are composed of spacer arms and anionic centers; the spacer arm provides flexibility to the oligomeric ethylene and/or ethylene oxide (EO) units, while the anionic centers, commonly carboxylate and sulfonyl imide, are responsible for the interaction with Li<sup>+</sup>. However, there exist problems with the utilization of these anionic centers, first, the generation of TFSI<sup>-</sup> anions in sulfonyl imides and their corrosive nature towards the Al cathode current collector, and second, but equally important, the feasibility of the anionic center to release the lithium ions from the carboxylate groups which limits on the ionic conductivity <sup>9</sup>.

SLICPE may have larger electrochemical stability windows and near-unit lithium transference numbers; however, the intrinsic ionic conductivity remains below the required values for use in LIB (e.g., cross-linked lithium electrolytes synthesized through poly(ethylene glycol) diacrylate with (4-styrenesulfonyl)(trifluoromethanesulfonyl)imide (STFSI) <sup>10</sup>, tetrakis(4-iodophenyl)borate with 1,4-diethynylbenzene <sup>11</sup>; and the blend tri(ethylene glycol) with lithium (4-styrenesulfonyl) (trifluoromethanesulfonyl)imide (LiSTFSI) and methoxy-polyethylene glycol acrylate <sup>6</sup>, obtaining with ionic conductivity values of  $3.2 \times 10^{-6} \text{ S cm}^{-1}$  at 55 °C,  $2.27 \times 10^{-4}$  at 30 °C and  $7.6 \times 10^{-6} \text{ S cm}^{-1}$  at 25 °C, respectively). Different approaches have been reported to improve the mobility of  $\text{Li}^+$  within the polyanionic environment and therefore, enhance their ionic conductivity <sup>6,7,9,10,12,13</sup>. Armand et al. presented a comprehensive review of the different SLICPE reported so far <sup>9</sup>, the review is mainly focused on the structural features that allow the immobilization of anions and dissociation of  $\text{Li}^+$  through the negative charge delocalization (electron density distribution). It has been suggested that polyanions with a high density of delocalized electrons allow increasing the dissociation rate of the ionic pair and increases the concentration of free  $\text{Li}^+$  in the polymer chains and, in consequence, the ionic conductivity <sup>13</sup>. Besides, in most cases the importance of the polymeric macromolecular structure is not considered, which has a greater impact on the dynamic properties of polymer chains and ergo, determine the electrochemical properties such as ionic conductivity, transference number and electrochemical stability <sup>2,5,11</sup>.

Many immobilized anions have been proposed, among them, the utilization of borate salts such as lithium bis(oxalate)borate (LiBOB) and lithium oxalldifluoroborate (LiODFBB) for the preparation of liquid and polymeric electrolytes has attracted attention due to their excellent ionic conduction properties <sup>12,14</sup>, and high thermal and electrochemical stability <sup>15</sup>.

In addition, their decomposition products favor the formation of a homogeneous solid electrolytic interface (SEI), from their decomposition products <sup>9</sup>, and the utilization of boron causes a very low environmental impact <sup>1,16</sup>.

Recently a new class of SLICPE based on borate groups reticulated by an ethoxy chain was synthesized from lithium tetramethoxy borate  $LiB(OCH_3)_4$  and PEG chains by a simple polycondensation method <sup>17</sup>; in these polymers, the electrochemical properties were modulated by the length of the PEG chains between the borate groups <sup>8</sup>. It was demonstrated that the increasing length of the PEG chains, allows higher dynamic movement of the PEG segments; therefore, the mobility of  $Li^+$  is faster and decreases the activation energy for the ionic conduction process. However, the relatively low ionic conductivity values,  $(4.95 \pm 0.05) \times 10^{-6} \text{ S cm}^{-1}$  at 25 °C in the best conditions, are related to the low  $Li^+$  concentration and the poor mobility for fully cross-linked polymeric matrix, and the low delocalized electron density in symmetrical borate groups which induces a high  $Li^+-BO_4^-$  interaction energy and limits the  $Li^+$  mobility <sup>18,19</sup>.

In this sense, it is necessary to propose new design and synthesis strategies that allow increasing the polymer intrinsic ionic conductivity; either, by allowing higher mobility of the polymer chains or by increasing the electronic delocalization in the borate groups <sup>20,21</sup>. In this work, the effect of these strategies on the electrochemical properties of SLICPE based on borate groups are described: firstly, we propose the increase of the polymer chain mobility by the formation of partially reticulated PEG400 chains; then, the electronic density within the immobilized anion is modulated by the incorporation of electron withdrawing groups (EWG) attached to the borate groups.

The SLICPE is synthesized by a conventional melting method while a detailed physicochemical characterization is carried to determine the changes associated with the partial reticulation and the effect of the EWGs. In addition, the effect of charge delocalization existing in the  $BO_4^-$  groups, associated to the strong electron withdrawing substituents, and their effect on the ionic conductivity in the SLICPE is described by a Density Functional Theory (DFT) analysis.

## 2. Materials and methods

### 2.1 Materials

Trimethyl borate (TMB) and lithium metal ( $Li^0$ ), were used to synthesize lithium tetramethoxy borate  $LiB(OCH_3)_4$  according to the procedure described by Barthel *et al.*<sup>17</sup>. Poly(ethylene glycol) 400 (PEG400) with an average  $M_n = 400$  g mol<sup>-1</sup> was dried by evaporation at 120 °C, using nitrogen flow and constant magnetic agitation for 2 hours in a ball flask. The alcohols, phenol ( $HOPh$ ) and 2,2,2-trifluoroethanol ( $HOCH_2CF_3$ ) were used as received. All reagents were obtained from Sigma-Aldrich.

### 2.2 Preparation of SLICPE based on asymmetrical $sp^3$ boron atoms

The reference SLICPE based on  $sp^3$  boron atoms fully cross-linked by PEG400 was synthesized by the reaction of Poly(ethylene glycol) 400 (PEG400) and ( $LiB(OCH_3)_4$ ) in a molar ratio 1 to 2 respectively, according to the method reported previously, and hereby named BO-EG<sup>8</sup>. SLICPE based on asymmetrical  $sp^3$  boron atoms partially cross-linked were synthesized, according to the proposed scheme shown in Figure 1, (75 mg, 5 mmol) of  $LiB(OCH_3)_4$  and 5 mmol of PEG400 were mixed under constant magnetic stirring in a 5 mL spherical flask under nitrogen atmosphere. The synthesis was carried out in two steps. In the

first step, reagents were homogenized by magnetic stirring at 100 °C for 1 hour; in the second step, the polycondensation reaction was carried out at 180 °C for 20 min with a nitrogen flow in order to remove the methanol generated during the reaction process and any traces of water from PEG reactants, the sample will be named BO-Me. Similar procedure was used to obtain partially cross-linked SLICPE based on asymmetrical  $sp^3$  boron atoms, incorporating organic EWG, as shown in Figure 1, by the reaction PEG400,  $(LiB(OCH_3)_4)$  and selected alcohols. In a 5 mL spherical flask under a nitrogen atmosphere and constant magnetic stirring were mixed (710 mg, 5 mmol) of  $(LiB(OCH_3)_4)$ , (1 g, 5 mmol) of PEG400 and 10 mmol of alcohols; (940 mg, phenol) and (0.72 ml, 2,2,2-trifluoroethanol) to obtain the samples BO-Ph and BO-CCF<sub>3</sub> respectively.

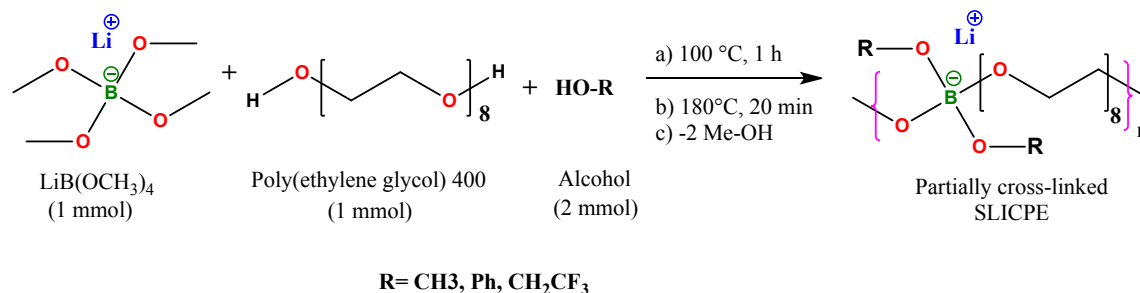


Figure 1. The synthesis scheme of asymmetrical SLICPE based on  $sp^3$  boron atoms, based on the incorporation of organic electron attractors groups (-R).

### 2.3 Physicochemical properties

Solid state  $^{13}\text{C}$ ,  $^{11}\text{B}$ , and  $^7\text{Li}$  NMR solid state spectra were acquired at room temperature, on MAS conditions at (6 kHz) and HPDEC technique, respectively, in a Bruker AVANCE-II 300 NMR spectrometer. Boric acid was used as a reference to the  $^{11}\text{B}$  chemical shift (chemical shifts are reported in ppm). Differential scanning calorimetry (DSC) was carried

out in a DSC 2500 TA instrument, with a nitrogen flow of  $50 \text{ cm}^3 \text{ min}^{-1}$ , at a heating rate of  $10 \text{ }^\circ\text{C min}^{-1}$  (first scan), to determine the melting point temperature  $T_m$  and glass transition temperature  $T_g$  of the samples, an amplitude of  $\pm 1.06 \text{ }^\circ\text{C}$  and a period of 40 s in the temperature range from -90 to  $130 \text{ }^\circ\text{C}$ .

## 2.4 Ionic conductivity measurements

**Ionic conductivity** ( $\sigma_{DC}$ ) was calculated from electrochemical impedance spectroscopy (EIS) measurements. A symmetrical stainless steel/polymer electrolyte/stainless steel cell was assembled. The distance between the electrodes ( $L$ ) was kept fixed at  $\sim 0.1 \text{ cm}$ , using a teflon spacer ring with an inner area ( $A$ ) of  $0.24 \text{ cm}^2$ . A Multi-Potentiostat/Galvanostat VMP3 from Bio-Logic Science Instruments was used for EIS measurements applying a 10 mV amplitude perturbation in the 1 MHz to 10 mHz frequency range at open circuit potential (OCP) conditions. The fits of the experimental results were carried out using a nonlinear regression algorithm implemented in the Zview® software. The ohmic resistance ( $R_0$ ) of the sample, estimated from the Nyquist plot at the low frequency end of the semicircle, was used to calculate the ionic conductivity ( $\sigma$ ) using the following equation:

$$\sigma = \frac{L}{(A R_0)}, \text{ (Scm}^{-1}\text{)} \quad (1)$$

Measurements were carried out from 10 to  $90 \text{ }^\circ\text{C}$  at every  $10 \text{ }^\circ\text{C}$  interval; the temperature was controlled, using a circulator coupled to a temperature bath (SEV, FC-10). Cells were allowed to reach thermal equilibrium for at least 1.5 h for their first test, and 30 min for the following tests <sup>22</sup>.

## 2.5 Computational Study

All electronic structure calculations were performed within the framework of Density Functional Theory (DFT) using the Vienna ab initio simulation package version 5.3.3<sup>23–25</sup>, the calculations were performed with the Perdew-Burke-Ernzerhof (PBE) exchange-correlation functional<sup>26</sup>. Core and valence electrons were treated with the PAW method<sup>27,28</sup> and expanded in plane waves with a cutoff energy of 500 eV. The polymer was simulated as a single chain polymer with different pendant groups and relaxed in a cell in such a way to avoid artificial self-interactions, i.e. at least 6 Å separated from any periodic image. The first Brillouin zone was sampled using 3x1x3 Monkhorst-Pack set of k-point meshes. For atomic charges, the Bader method was utilized as implemented in reference<sup>29</sup>. Activation energy for Li<sup>+</sup> hopping was calculated using the climbing Nudged Elastic Band Method (cNEB)<sup>30</sup>, linear interpolation between the reactant and product states of each elementary step (a total of 5 images) are used as initial guesses for the reaction coordinate, and the individual images are then optimized with the cNEB algorithm, the force criterion for structure optimization and cNEB search was set to 0.02 eV Å<sup>-1</sup>. Ab-initio Molecular Dynamics were utilized to analyze Li<sup>+</sup> mobility, the (TVN) ensemble was used at 300K with a time step of 1 fs.

## 3. Results and discussion

In this work, the conductivity of the polymer is expected to be modified by two effects: the partial crosslinking process and the effect of the EWGs. In order to determine the extent of the effect of the EWG, a preliminar DFT analysis was performed for a  $BO_4^-$ -Li<sup>+</sup> salt in vacuum substituted with -CH<sub>3</sub> and -CF<sub>3</sub> groups, Figure 2. The differences on the structure and properties of the salt caused by the substituents indicates that effectively, a more electron withdrawing group (The group having more electronegative atoms, -CF<sub>3</sub> vs -CH<sub>3</sub>) has an

important effect on the decrease of the charge associated to the  $BO_4^-$  group, increasing the O-Li<sup>+</sup> distance by  $\sim 0.06$  Å and lowering the Li<sup>+</sup>- $BO_4^-$  interaction Energy by  $\sim 1$  eV, as shown in Figure 2. Therefore, it is expected that the conduction properties can be modulated by the presence of highly EWGs in the partially cross-linked polymers. Later, in this work, three substituents were utilized to modify the SLICPE properties, -CH<sub>3</sub> (BO-Me), -CH<sub>2</sub>CF<sub>3</sub> (BO-CCF<sub>3</sub>) and -Ph (BO-Ph). The changes on the substituents are expected not only to have an effect as EWGs, but also possess steric and structure type influence, therefore, determining the overall physicochemical properties of the polymer electrolyte. Thus, in the following, the molecular structures of the synthesized SLICPE are corroborated by solid state NMR, the changes in the mobility properties of the polymeric matrix are evaluated by the glass transition temperature ( $T_g$ ) obtained by DSC measurements and the ionic conductivity is obtained by the EIS technique.

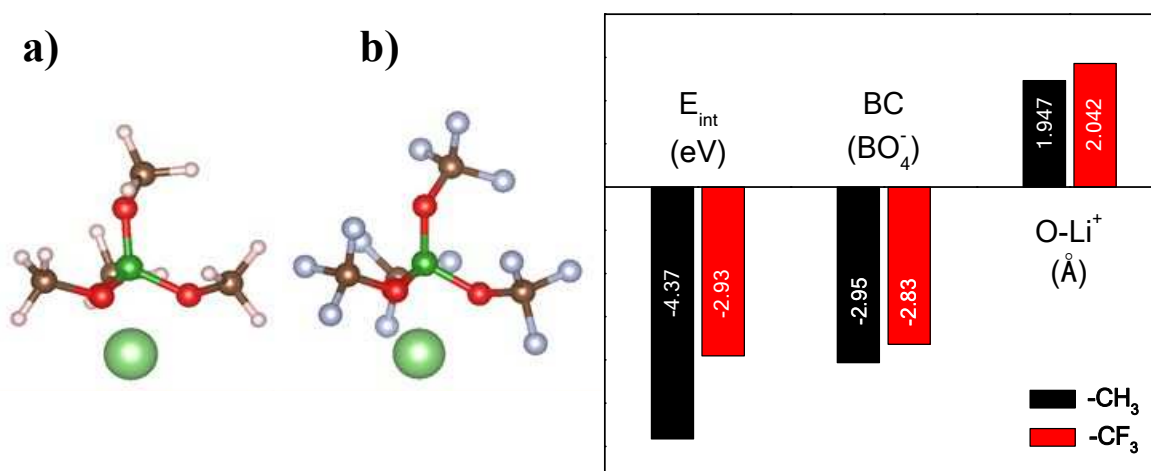


Figure 2. Representation of the molecular structure of borate groups substituted with different electroattractor groups, obtained by DFT calculations; a) -CH<sub>3</sub>, b) -CF<sub>3</sub> and c) calculated parameters for the different structures;  $BO_4^-$  - Li<sup>+</sup> interaction Energy ( $E_{int}$ ), Bader Charges\* (BC) and O-Li<sup>+</sup> distance. Boron, lithium, oxygen, carbon, hydrogen and fluorine are

represented by small green, big green, red, brown, white and blue spheres, respectively. \*In this work, Bader Charges are reported as the difference in the number of electrons within the Bader volume and the valence electrons of the atom, in this way the units are excess electrons per molecule.

### 3.1 NMR solid state analysis of SLICPE based asymmetrical $sp^3$ boron atoms

The chemical structure of the reference SLICPE “BO-EG” prepared by polycondensation was confirmed by solid-state  $^1\text{H}$  and  $^{13}\text{C}$  NMR spectroscopy, the spectra obtained at 5kHz are shown in figures 3ai and 3bi, respectively. The sharp signal with a chemical shift at 3.7 ppm in the NMR spectrum of  $^1\text{H}$  for BO-EG is assigned to the protons of the PEG400 ethoxy chain between borate groups (Figure 3ai). While the signals observed in the  $^{13}\text{C}$  NMR spectrum, with chemical displacements of 60.5, 71.5 and 69.6 ppm are assigned to the carbon atoms ( $\alpha$ )  $\sim\text{BO}-\text{C}-\text{C}-\text{O}\sim$ , ( $\beta$ )  $\sim\text{BO}-\text{C}-\text{C}-\text{O}\sim$  and ( $\gamma$ )  $\sim\text{O}-\text{C}-\text{C}-\text{O}\sim$ , respectively <sup>1,8</sup>.

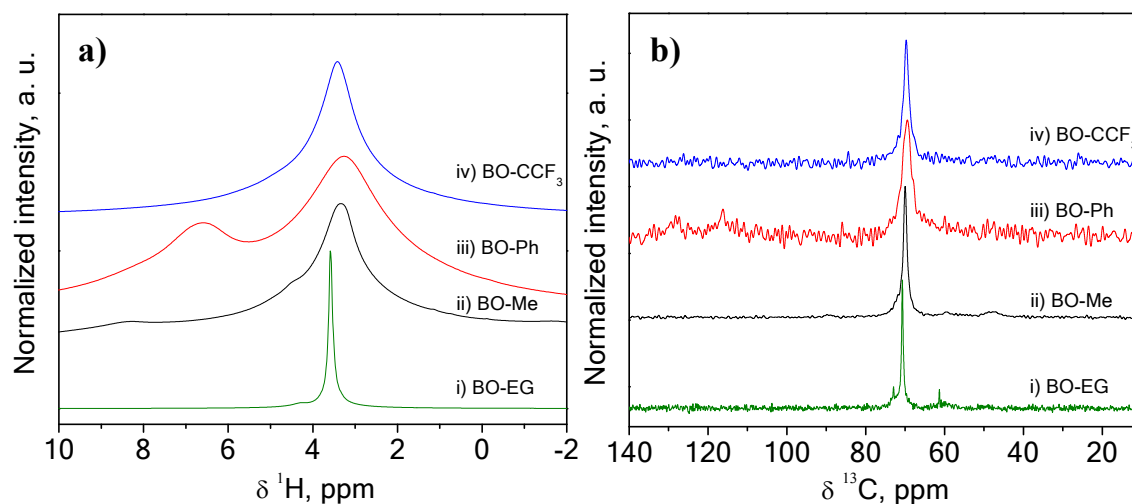


Figure 3. a)  $^1\text{H}$  and b)  $^{13}\text{C}$  NMR/MAS spectra of synthesized SLICPE; i) BO-EG, ii) BO-Me, iii) BO-Ph and iv) BO-CCF<sub>3</sub>.

The  $^1\text{H}$  and  $^{13}\text{C}$  NMR MAS spectra for partially reticulated SLICPE BO-Me, BO-Ph and BO-CCF<sub>3</sub> were acquired 3 kHz, the significant increase in the width of the signal with respect to BO-EG is associated to the effect of the decreased rotation frequency during spectra acquisition. In the  $^1\text{H}$  NMR spectra of the samples, a relatively large peak associated to methylene protons adjacent to the oxygen atom of the ether predominates ( $\gamma$ )  $\sim\text{CO-C-C-OC}\sim$  with a chemical displacement of 3.3 ppm (Figure 3a). In addition, in the analysis of the NMR spectra of  $^1\text{H}$  and  $^{13}\text{C}$  for partially cross-linked SLICPE new characteristic signals are identified. For the SLICPE BO-Me, the  $^{13}\text{C}$  NMR spectra presents a signal at 4.2 ppm associated with carbon ( $\beta$ )  $\sim\text{BO-C-C-O}\sim$ , which indicates the covalent addition of PEG400 chains on the borate groups; on the other hand. Although the width of the signal of  $^1\text{H}$  NMR spectra of the ethoxy chain does not permit clearly identify the signal associated with the carbon protons ( $\alpha$ )  $\sim\text{BO-CH}_2\text{-CH}_2\text{-O}\sim$  and  $\sim\text{BO-CH}_3$ , those should be located approximately in 3.65 and 3.39 ppm respectively (Figure 3aii); moreover the signals appearing in the  $^{13}\text{C}$  NMR spectra with chemical shift  $\sim 60\text{-}40$  ppm were assigned to the carbon atoms of the substituted methoxy groups on the boron atoms with  $\text{sp}^3$  coordination  $\text{BO-CH}_3$  (Figure 3bii). The  $^1\text{H}$  NMR spectrum obtained for BO-Ph presents a wide resonance signal with chemical displacement between 6 and 7.5 ppm characteristic of the protons of the aromatic groups (figure 3aiii), whose presence is corroborated by the appearance of signals with chemical displacement in 120 and 130 ppm in the corresponding  $^{13}\text{C}$  NMR spectrum (Figure 3biii). For the NMR spectra obtained for BO-CCF<sub>3</sub>, it is not possible to identify the characteristic signals of protons and methylene carbons  $\text{-CH}_2\text{-}$  of the substituted 2,2,2-trifluoroethanol ( $\text{HOCH}_2\text{CF}_3$ ), due to the similarity of their chemical displacements with the characteristic displacements of the methylene groups of the PEG400 ethoxy chain (Figures 3aiv and 3biv).

The borate structures of the SLICPE were investigated by solid-state  $^{11}\text{B}$  and  $^7\text{Li}$  NMR CP/MAS (Figure 4). The  $^{11}\text{B}$  NMR CP/MAS spectrum obtained for the BO-EG reference SLICPE shows a single relatively narrow signal in the range 7-9 ppm (Figure 4ai), which indicates a highly symmetrical arrangement of the four oxygen atoms in the tetrahedron formed by  $\text{BO}_4^-$  groups and suggests that all boron atoms retain  $\text{sp}^3$  coordination. In addition, the  $^7\text{Li}$  NMR CP/MAS spectrum obtained for BO-EG exhibited a narrow signal at 0.2 ppm, characteristic of polymeric electrolytes with low  $\text{Li}^+$  concentration (Figure 4bi). The  $^{11}\text{B}$  and  $^7\text{Li}$  signals obtained for BO-EG corresponds to the totally cross-linked polymer without functional groups, therefore they will serve as a reference point for the changes arising from the formation of partially cross-linked polymer and the effect of the substituent groups.

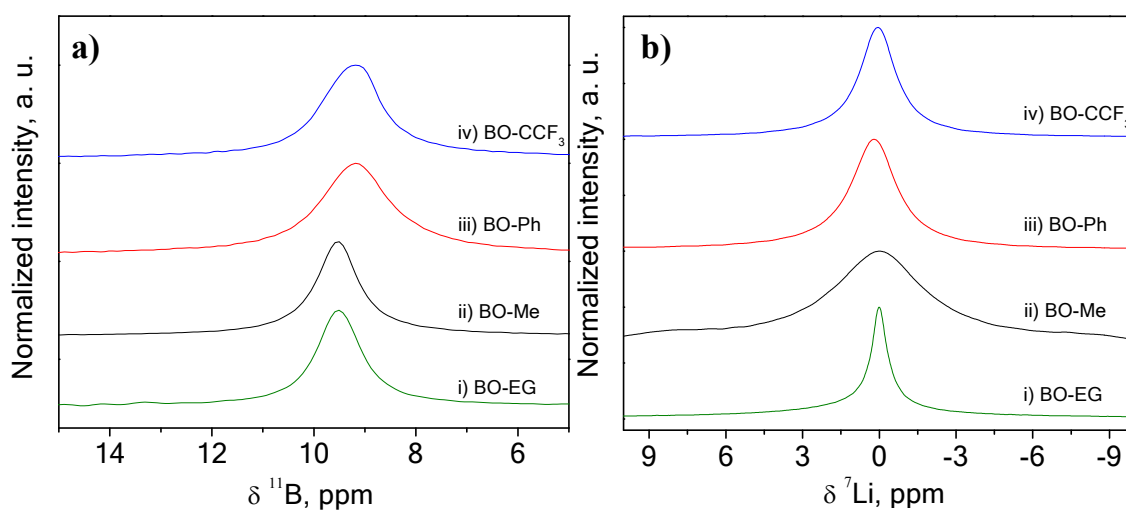


Figure 4. a)  $^{11}\text{B}$  and b)  $^7\text{Li}$  NMR CP/MAS spectra of synthesized boron-based SLICPE; i) BO-EG, ii) BO-Me, iii) BO-Ph and iv) BO-CCF<sub>3</sub>.

The solid-state  $^{11}\text{B}$  NMR spectrum CP/MAS, obtained for the partially cross-linked SLICPE BO-Me, BO-Ph and BO-CCF<sub>3</sub> shows a single narrow signal in the range 7-11 ppm

(Figure 4a-iv), characteristic of  $sp^3$ -coordinated boron atoms, demonstrating the effectiveness of the synthesis method; otherwise, the presence of multiple signals could indicate the formation of several coordination structures, including  $sp^2$ -coordinated boron atoms. The similarity of the width and the chemical displacement of the signals in the  $^{11}\text{B}$  spectra obtained for BO-EG and BO-Me suggests that the chemical environment of the borate groups does not perceive significant changes. On the other hand, the displacement towards higher fields of the signals associated to the borate groups in the spectra for the SLICPE BO-Ph and BO-CCF<sub>3</sub> is directly related to the decrease of the  $^{11}\text{B}$  RMN chemical shift obtained for the SLICPE BO-Ph and BO-CCF<sub>3</sub> and is probably related to lower interaction energy between  $\text{Li}^+$  and the borate groups (Figure 4a-iv).

In order to get a direct interpretation of the NMR spectra, DFT analysis of single polymer chain modified with the different substituents was performed. The single chain was substituted with two  $-\text{CH}_3$ ,  $-\text{CH}_2\text{CF}_3$  and  $-\text{Ph}$  groups; an additional system substituted with  $-\text{CF}_3$  was also calculated for comparison. The systems were allowed relaxing until the lowest energy configuration was obtained, Bader Charges,  $\text{Li}^+$ -O distances and interaction energy ( $E_{\text{int}} = E_{\text{polym-Li}^+} - E_{\text{polym}} - E_{\text{Li}^+}$ ) were calculated for all systems (Table 1).

Table 1. DFT calculated properties of single chain  $\text{BO}_4^-$  partially crosslinked with the different substituent groups. Interaction energy between  $\text{BO}_4^-$  and  $\text{Li}^+$  ( $E_{\text{int}}$ ) and Bader charge (BC).

|                       | Substituent group ( $-\text{R}$ ) |                |                           |              |
|-----------------------|-----------------------------------|----------------|---------------------------|--------------|
|                       | $-\text{CH}_3$                    | $-\text{CF}_3$ | $-\text{CH}_2\text{CF}_3$ | $-\text{Ph}$ |
| $E_{\text{int}}$ (eV) | -5.06                             | -4.88          | -5.13                     | -4.68        |

|                                                |       |       |       |       |
|------------------------------------------------|-------|-------|-------|-------|
| (BO <sub>4</sub> <sup>-</sup> ) (Bader Charge) |       |       |       |       |
| (Excess electron)                              | -2.97 | -2.87 | -2.95 | -2.93 |
| O-Li <sup>+</sup> (Å)                          | 1.792 | 1.814 | 1.807 | 1.801 |

It is worth noting that in comparison to -CH<sub>3</sub> substituents most of the electro-attractors groups effectively diminish the interaction energy of Li<sup>+</sup> with the borate group, among them, the aromatic group decreases drastically the interaction energy. However, not all substituents show the same trend, for instance, the -CH<sub>2</sub>CF<sub>3</sub> group increases the interaction energy, this behavior is related to the mobility of the long -CH<sub>2</sub>CF<sub>3</sub> chain, capable to intra interact with the Li<sup>+</sup> increasing the overall interaction energy. The influence of the electron withdrawing group is also observed on the Bader Charge of the borate group which also diminishes for all EWG in comparison to -CH<sub>3</sub>. In the same line, the Li<sup>+</sup>-O distance is slightly increased, indicating a direct effect of the lower interaction energy. With the results obtained so far, it is not possible to obtain a direct correlation with the experimental RMN shifts; however, it should be noted that the real polymeric structure is very complex, containing amorphous and crystalline domains in which the mobility and interactions between them can give rise to totally different properties. Obviously, a complete representation of all the properties of a polymer is highly computationally demanding and, therefore, out of the scope of this paper.

In order to get a more realistic representation of the polymer, a second model was constructed in which two linear chains with the side EWGs were directed towards the other chain in different positions (Figure 5). Thus, any extra energy is a direct consequence of the interaction of the substituent group with a section of the polymer or otherwise, Li<sup>+</sup>. Neither

the -CH<sub>3</sub> nor the -Ph groups promote an important increment on the interaction energy (0.02 eV); however, only the -CH<sub>2</sub>CF<sub>3</sub> substituent increases the energy of the system by 0.5 eV, only in the scenario in which the fluorine atoms point directly to the Li<sup>+</sup> on the neighboring chain. This extra interaction also leads to an increase in the Li<sup>+</sup>-O distance from 1.807 to 1.857 Å, which indicates a lower interaction of the Li<sup>+</sup> with the borate group and probably could be related to higher Li<sup>+</sup> mobility.

Therefore, the influence of the EWG may have two components: 1) The effect of the substituent group on the borate (to which is directly bonded) by the direct effect of delocalization of charge allowing a lower Li<sup>+</sup>-O interaction energy, and 2) the effect of the substituent group by the direct interaction of the end of the group (or a part of it) with Li<sup>+</sup> in another polymer chain. Therefore, in this work, the -CH<sub>3</sub> does not present any effect on the borate; the aryl group, on the other hand, due to its size and relatively limited mobility exclusively presents the direct effect of charge delocalization by the electron withdrawing group directly on the BO<sub>4</sub><sup>-</sup> moiety and; finally, the -CH<sub>2</sub>CF<sub>3</sub> besides its electro-attracting effect, it can also contribute to the Li<sup>+</sup> mobility by the direct interaction of the fluorine atoms with Li<sup>+</sup> (Figure 5).

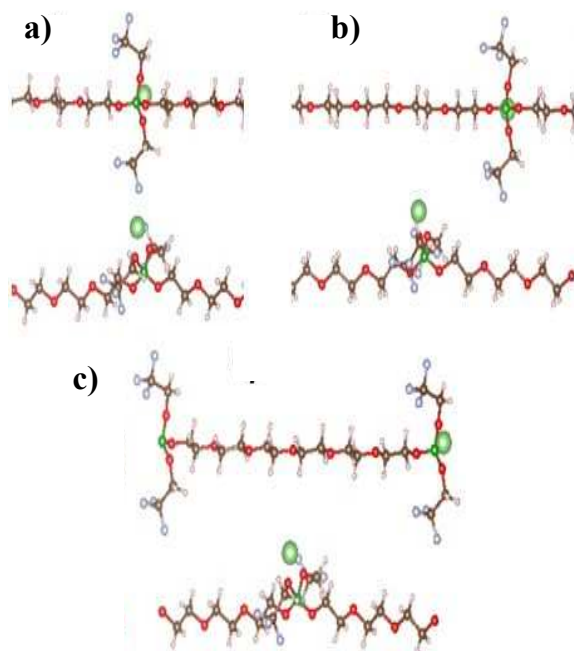


Figure 5. Schematic representation of the possible interactions between polymer pairs a) Direct interaction between the substituent group of one polymer chain and  $\text{Li}^+$  of the other chain, b) and c)  $\sim 3.0$  and  $\sim 6.0$  Å separation between the substituent group and  $\text{Li}^+$  in the other polymer chain, respectively.

So, the changes in the  $^{11}\text{B}$  NMR spectra can be associated with the effect of the substituent groups on the  $\text{BO}_4^-$  anions. The  $^7\text{Li}$  NMR spectra of the asymmetric SLICPE, also shows important changes with respect to the BO-BEG. The decrease in the width of the  $^7\text{Li}$  RMN signals obtained for the BO-Ph and BO-CCF<sub>3</sub> SLICPE, with respect to BO-Me, is probably a direct consequence of the increment on the mobility of  $\text{Li}^+$  through the structure of the main polymeric structure (Figure 4biii-iv), corroborating that the decrease in the interaction energy between  $\text{Li}^+$  and borate groups of the partially reticulated SLICPE. In order to corroborate the mobility of  $\text{Li}^+$  along the chain, activation barriers for  $\text{Li}^+$  hopping

from the borate group to the nearest ethoxy group were calculated by the Nudged Elastic Band method (Figure 6).

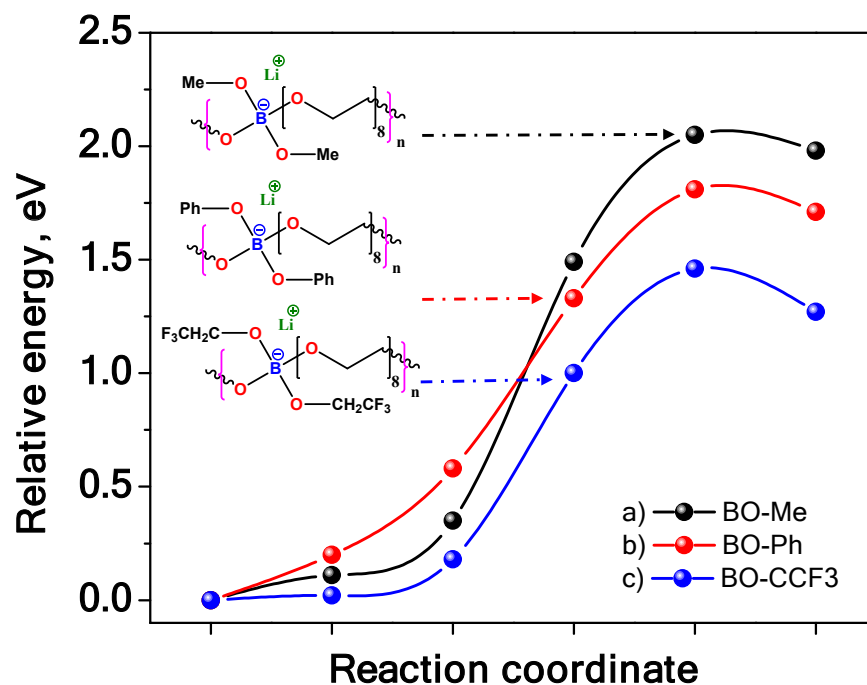


Figure 6. DFT calculated activation barriers for  $\text{Li}^+$  hopping from the borate to the adjacent ethoxy group on the same polymer chain.

The process for  $\text{Li}^+$  hopping from the  $\text{BO}_4^-$  group to the adjacent ethoxy group is an uphill process, indicating that  $\text{Li}^+$  is hopping from a highly stable site to less stable one. The influence of the EWG is also readily observed since the activation energy for the process is different in each of them. The activation energy for  $\text{Li}^+$  hopping for the  $-\text{CH}_3$  substituted polymer is the highest, decreasing slightly for  $-\text{BO-Ph}$  and drastically for  $\text{BO-CCF}_3$ . This trend perfectly matches the effects of the substituents on the interaction energy of  $\text{Li}^+$  with the  $\text{BO}_4^-$  group. It should be noted that this hopping mechanism is not the only one possible, there exist several possibilities: hopping mediated by chain movement, mediated by the interactions with neighboring polymer chains or other crystalline or amorphous regions of

the polymer. However, these preliminary results allow effectively correlating the influence of the groups on the interaction energy and mobility observed by the shifts on the NMR spectra and widening of the signals of  $^7\text{Li}$  NMR.

In this section, it has been demonstrated that the electron density of  $\text{BO}_4^-$  groups is a direct function of the electron withdrawing ability of the substituent organic groups covalently linked to boron atoms, since the greater the electron withdrawing ability of the organic substituent, the lower the electron density of  $\text{BO}_4^-$  groups and their interaction with  $\text{Li}^+$  is lower, favoring the  $\text{Li}^+$  mobility through the polymeric matrices.

However, there still some unresolved properties. For instance, the increase of the signal width of the  $^7\text{Li}$  NMR spectrum obtained for BO-Me with respect to BO-EG is characteristic of a more concentrated polymer electrolyte and vitreous states with the relatively static environment, in which the dissociation of  $\text{Li}^+$  decreases and its interaction energy with the polymer chains increases with the concentration of  $\text{Li}^+$  <sup>31</sup>. Also, the width increment of these signals with respect to BO-Me is associated to the formation of borate groups with multiple structural configurations, generated by the random substitution of the organic electro-attractor groups on the boron atoms.

### 3.2 DSC analysis of SLICPE based asymmetrical $\text{sp}^3$ boron atoms

Differential scanning calorimetry was used to determine the glass transition temperature ( $T_g$ ) of the synthesized SLICPE (Figure 7). According to the DSC plots, a  $T_g$  transition is observed at  $-64\text{ }^\circ\text{C}$ ,  $-46\text{ }^\circ\text{C}$ ,  $-62\text{ }^\circ\text{C}$  and  $-68\text{ }^\circ\text{C}$  for samples BO-EG, BO-Me, BO-Ph, and BO- $\text{CCF}_3$ , respectively, and suggest high free volume values available for the mobility of polymer chains (Figure 7ii-iv), which is reasonable considering the size of the substitute

electroattractor groups and the ease of obtaining homogeneous matrices, either cross-linked or partially cross-linked. Additionally, BO-Ph sample calorimetry analysis show a transition signal at  $-39^{\circ}\text{C}$  which can be associated with the glass transition of compounds formed from borate groups with multiple structural configurations, due to the random substitution of the electro-attractor aryl groups on the boron atoms during the synthesis process.

Contrary to the expectations, the  $T_g$  values obtained for the partially cross-linked SLICPE do not show a significant decrease with respect to the  $-64^{\circ}\text{C}$ , characteristic value of the fully cross-linked BO-EG SLICPE (Figure 7i), this anomalous behavior can be clearly explained by considering the loss of mobility of the PEG chains generated by the increased concentration of ionic pairs and the interaction energy between  $\text{Li}^+$  and the regions with high electron density in the polymeric matrix <sup>32</sup>.

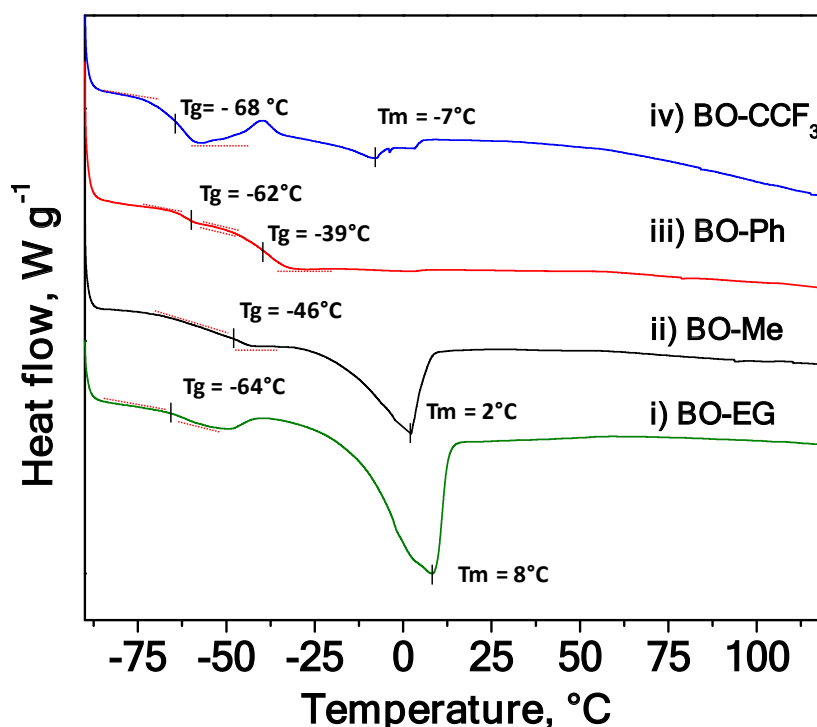


Figure 7. DSC spectra of synthesized SLICPE; i) BO-EG, ii) BO-Me, iii) BO-Ph and iv) BO-CCF<sub>3</sub>. The temperature regions used for  $T_g$  valuation, and the  $T_g$  and  $T_m$  values are indicated in the figure.

On the other hand, the EWG linked to the  $BO_4^-$  can give rise to the formation of short-range electrostatic interactions capable of stiffening the polymeric matrix<sup>33</sup>, either by the interaction between two or more groups substituted in different boron atoms or substituent groups with other regions of the polymeric matrix<sup>14,34</sup>. These multiple interactions can affect the mobility of the polymeric matrix of the SLICPE, which, in consequence, can have predominant effects on the mobility of the  $Li^+$ , mainly when the predominant ionic conduction mechanism is favored by the PEG chains movement.

These semi-crystalline domains formed by the different short-range electrostatic interactions between polar regions of the PEG400,  $Li^+$  chains, and electro-attractor groups do not show an apparent correlation with the nature of the molecular structure of the substitute electro-attractor groups, since the BO-EG, BO-Me, and BO-CCF<sub>3</sub> SLICPE exhibit an endothermic signal at 8 °C, 2°C and -7 °C, respectively associated with the melting process of these domains (Figure 7i, 7ii and 7iv), BO-Ph sample does not have an evident melting process (Figure 7iii). The melting temperature ( $T_m$ ) for partially cross-linked SLICPE are below the typical values of melting processes of reference SLICPE BO-EG and polyethylene oxide (PEO), 8 °C and 65 °C respectively<sup>35,36</sup>.

These  $T_m$  values suggest that the energy required to reach the disorder of polymer chains and the melting of semi-crystalline domains formed by electrostatic interactions is lower in partially cross-linked SLICPE. Therefore, it is expected that for temperatures above

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3 *T<sub>m</sub>* the dynamics of the movement of the polymeric chains and Li<sup>+</sup> are favored, overcoming  
4 the limitations of structural rigidity; therefore, above this temperature the real effect of the  
5 different electro-attractor substituents on the conductivity of Li<sup>+</sup> may become evident due to  
6 the variations of the electronic density of the asymmetric borate groups.  
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### 15 **3.3 Ionic conductivity analysis of SLICPE based asymmetrical sp<sup>3</sup> boron atoms**

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17 The variation of ionic conductivity as a function of temperature obtained for SLICPE based  
18 borate groups partially cross-linked by PEG400 bridges is shown in Figure 8a. SLICPE BO-  
19 Me, BO-CCF<sub>3</sub> and BO-Ph exhibit ionic conductivity values of  $(1.03 \pm 0.04) \times 10^{-6}$ ,  $(3.48 \pm$   
20  $0.04) \times 10^{-6}$  and  $(4.02 \pm 0.04) \times 10^{-6} \text{ Scm}^{-1}$  at 30 °C, respectively (Figure 8a-iv). The ionic  
21 conductivity values of  $(4.95 \pm 0.02) \times 10^{-6} \text{ Scm}^{-1}$  at 30 °C obtained for the fully reticulated BO-  
22 EG SLICPE (reference SLICPE, Figure 8ai) are opposite to those expected according to the  
23 initial hypothesis, where it is proposed that increasing the O/Li<sup>+</sup> ratio to 10 and increasing  
24 the free volume for the mobility of the polymer chains generated for the partial cross-linking  
25 of the borate groups through the decrease of the crosslinking agent PEG400 in the SLICPE  
26 matrix.  
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40 This apparently anomalous behavior is associated with the possible formation of  
41 secondary semi-crystalline domains or glasses with different boron/oxygen stoichiometric  
42 ratios <sup>8,37</sup>, derived from partial polycondensation reactions between LiB(OCH<sub>3</sub>)<sub>4</sub>, PEG400  
43 and the used alcohols as a source of electro-attractors; this is also related to the increased  
44 structural rigidity generated by semi-crystalline domains formed by short-range electrostatic  
45 interactions between polar groups of the PEG400, Li<sup>+</sup> chains and organic electro-attractors  
46 covalently linked to the borate groups. This behavior suggests that the effect of structural  
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properties is dominant over the electro-attractor effect of the substituent groups in  $BO_4^-$  groups, on the ionic conduction properties in the region close to the melting temperatures.

Ab initio Molecular Dynamic (AIMD) were also performed in order to detect other geometries and possible  $Li^+$  hopping from one site to another. For the duration of the AIMD (10 ps) at 300K, no  $Li^+$  hopping was detected; however, additional interesting results were obtained as described below. The  $Li^+$ -O distance was analyzed during the duration of the simulation, the average distance results in the specific values 1.825, 1.845 and 1.83 Å for BO-Me, BO-CCF<sub>3</sub>, BO-Ph, which indirectly relates with the interaction energy and mobility of  $Li^+$ , those values agree with the calculations and experiments analyzed above. During the simulation, all structures transit between different short time configurations with shorter and larger bond distances; however, the BO-Me structure transits over a configuration consisting of B-O-Li-O-B arrangements which once it is reached, it persists for the rest of the simulation. In this specific configuration, a larger O- $Li^+$  bond length is observed; however, once this configuration is reached, the energy needed to move the  $Li^+$  ion is higher, thus the mobility of the  $Li^+$  is diminished. For BO-CF<sub>3</sub> (results not shown), B-O-Li-O-B configuration is also obtained resulting in similar low conductivity. Therefore, for short substituents, the specific interaction between the chains can lead to  $Li^+$  trapping resulting in decreased  $Li^+$  mobility.

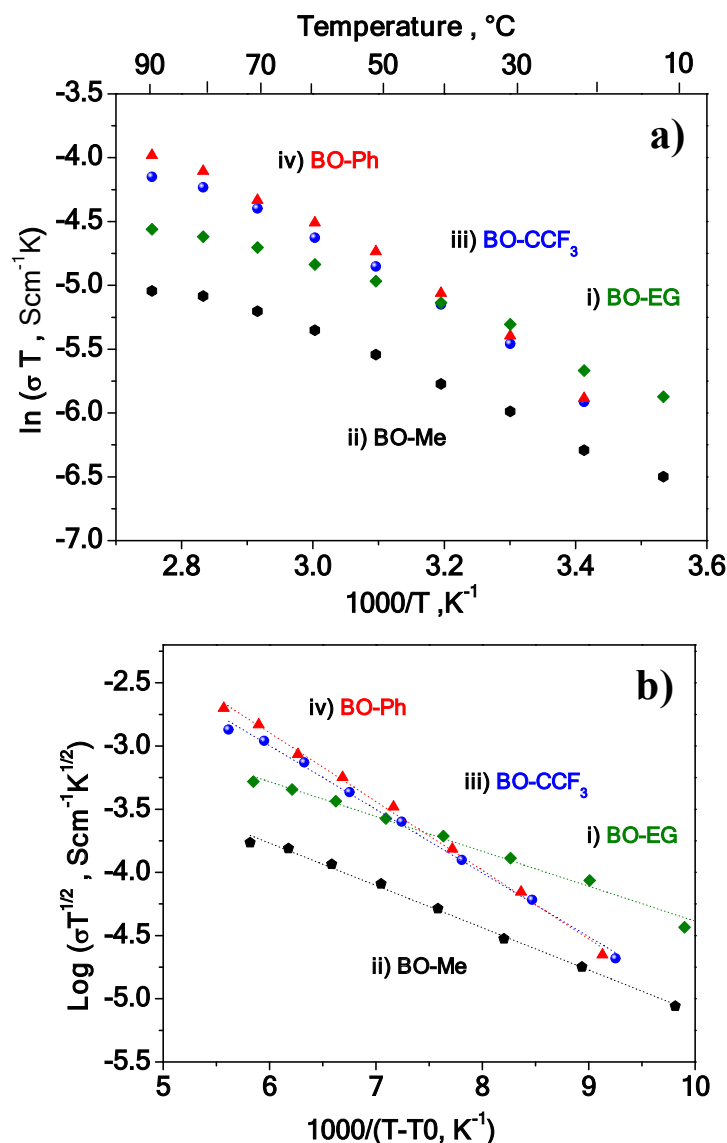


Figure 8. Ionic conductivity as temperature function for synthesized SLICPE; a) Arrhenius plot and b) VTF plot; i) BO-EG, ii) BO-Me, iii) BO-Ph and iv) BO-CCF<sub>3</sub>.

On the other hand, the increase in ionic conductivity values observed in SLICPE partially cross-linked BO-CCF<sub>3</sub> and BO-Ph (Figure 8a iii-iv), in the high-temperature region (above 40 °C) suggest that the increase in temperature in SLICPE generates the energetic conditions and dynamic disorder necessary to overcome the structural stiffness limitations. Therefore, the ionic conductivity values in the high temperature region show the relationship between the electron withdrawing capacity of the substituting EWG and the ionic

conductivity (Figure 8a-iii-iv), which indicates that the greater the electron withdrawing capacity of the substituting groups, the greater the ionic conductivity, which can be directly related to the decrease in the electron density in  $BO_4^-$  groups and consequently the decrease in the interaction energy between  $BO_4^-$  and  $Li^+$  groups, which favors ionic mobility.

The ionic conductivity measurements obtained in the high-temperature region (above 40°C) allow to the semi-crystalline domains to be melted in the partially cross-linked BO-CCF<sub>3</sub> and BO-Ph SLICPE and diminish the effect of short-range electrostatic interactions on ionic conductivity values by obtaining ionic conductivity values of  $(1.40 \pm 0.04) \times 10^{-5}$  and  $(1.84 \pm 0.04) \times 10^{-5} \text{ Scm}^{-1}$  at 50 °C, respectively (Figure 8a-iii-iv), surpassing the value of the fully cross-linked BEG8 SLICPE of  $(1.07 \pm 0.04) \times 10^{-6} \text{ Scm}^{-1}$  at 50 °C. However, the ionic conductivity value of  $(2.87 \pm 0.04) \times 10^{-6} \text{ Scm}^{-1}$  at 50 °C exhibited by the SLICPE BO-Me remains below the reference and may be associated with the nature and concentration of its semi-crystalline domains.

The activation pseudo-energy values of the partially reticulated SLICPE, calculated by means of the Vogel-Tamman-Fulcher (VTF) equation that relates ionic motion and mobility of polymer chains (Figure 8b-iv), determined by the glass transition temperature  $T_g$ , with values of 0.034, 0.064 and 0.063 eV, for SLICPE BO-Me, BO-Ph and BO-CCF<sub>3</sub>, respectively, are close to the pseudo activation energy value obtained for BO-EG of 0.032 eV and suggest that the preferential ionic conduction mechanism is by  $Li^+$  hopping through the oxygen atoms of the PEG400 chains between  $BO_4^-$  groups, favored by the mobility of the chains<sup>38–40</sup>. Furthermore, the values of pseudo-energy of activation for partially cross-linked SLICPE do not show sufficiently important changes to suppose that the preferential ionic transport mechanisms were modified with respect to the preferential ionic transport

mechanisms in fully cross-linked SLICPE and similarly the values of lithium transference numbers for partially cross-linked SLICPE are expected to remain close to the unit between 0.88 and 0.98 as reported for fully cross-linked SLICPE.

#### 4. Conclusions

The influence of some electro-attractor substituents attached to the immobilized anion in boron-based SLICPE has been studied by a computational/experimental approach. It has been determined that the chemical environment and  $\text{Li}^+$  mobility is indeed affected by the presence of substituents capable to delocalize the charge in the immobilized anion, thus modifying the  $\text{Li}^+$ -O distance and its interaction Energy. However, the mere presence of the substituents is not enough to increase the ionic conductivity, since the mobility of the chains is very limited at lower temperatures, only at temperatures above 40 °C, the effect of the substituents has a greater effect. Thus, in this study we have found that in order to improve the conductivity of boron base polymers, the mobility of the chains should be first improved, then the effect of these or other substituents can have a deeper effect.

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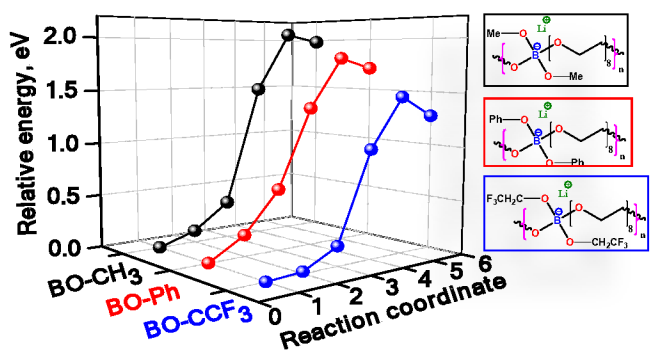
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